

Thomas Van Camp

(224) 281-3667 | thomas.j.vancamp@gmail.com | [linkedin.com/in/thomas-van-camp/](https://www.linkedin.com/in/thomas-van-camp/) | <https://tvc04.github.io/>

EDUCATION

University of Michigan, Ann Arbor, College of Engineering

August 2023 - May 2027

Major: Computer Science | Minor: Mathematics

GPA: 3.94 / 4.0

- **Activities:** Tau Beta Pi (Engineering Honor Society), Quantum Computing Club, Dean's Honor List, Michigan Hackers (Security Team), Michigan Ski and Snowboard Club (President)
- **Relevant Courses:** Architecting Hybrid Quantum-Classical Systems, Intro to Quantum Computing (PHYS 380), Intro to Quantum Computing (EECS 479), Intro to Quantum Information Technology, Intro to Computer Security, Applied Linear Algebra, Data Structures and Algorithms, Intro to Computer Organization, Foundations of Computer Science, Discrete Math

RESEARCH

University of Michigan, CAFQA Lab

August 2026 - Present

Undergraduate Research Assistant

- Exploring and improving PureMagic, a dynamic lattice-surgery compiler for fault-tolerant quantum computing, with a focus on resource-efficient scheduling of logical operations and magic-state cultivation
- Investigating compiler scheduling and resource-allocation techniques for surface-code quantum computing

Michigan State University, QCSL

May 2026 - July 2026

Undergraduate Research Assistant

- Selected as one of ten students of over 200 applicants to the NSF-sponsored MSU ACRES REU (Award #2349002)
- Conducted research under Dr. Ryan LaRose on efficient classical simulations of large-scale unitary cluster Jastrow circuits
- Created a pipeline to turn target molecules into quantum circuits, allowing for simulation using matrix product state methods
- Optimized scripts for GPU-based workflows on ICER's HPC that utilized Python's Qiskit, Quimb, pysef, and ffsim libraries
- Presented a poster and delivered an oral report of our research project findings at the 2026 Mid-SURE conference

PUBLICATIONS

Hrishikesh Belagali, Thomas Van Camp, R. Pradeep, Sourin Das, Namit Anand, Ryan LaRose, Efficient classical simulation of large-scale unitary cluster Jastrow circuits. In review Phys Rev Lett (2026). arXiv:2607.21337

RELEVANT WORK EXPERIENCE

AbbVie Inc., North Chicago, IL

May 2025 - August 2025

Software Engineer Intern, EPIC Team

- Created a Chrome Extension to promote efficient and corporate-compliant social media use
- Collaborated with communications professionals to ensure accurate LLM responses and intuitive UX
- Developed a Streamlit web app for media reporting and summaries using web API endpoints
- Presented projects at several group meetings utilizing live demonstrations and posters
 - Technologies: GitHub, Python (Streamlit), LLMs, Chrome Extension APIs, Web UI/UX

AbbVie Inc., North Chicago, IL

May 2024 - August 2024

Software Engineer Intern, Therapeutics Artificial Intelligence Team

- Created Flask APIs to auto-generate 10 sections of Periodic Safety Update Reports via web requests
- Achieved 91% success rate for engineered LLM prompts to automate medical error narrative classification
 - Technologies: GitHub, Python, Flask RESTful, LLMs (GPT-4o, Claude, GPT-3.5 Turbo)

AbbVie Inc., North Chicago, IL

June 2023 - August 2023

Cloud Engineer Intern, Scientific and Cloud Computing Group

- Created 3 scripts to maintain AWS cloud security and ensure compliance with updated corporate policies
- Deployed automated scripts via 200+ Jenkins pipelines to delete unapproved Identity Access Management (IAM) user keys, policies, and roles
 - Technologies: AWS IAM, AWS Cloud, Jenkins, GitHub, Python

AbbVie Inc., North Chicago, IL

June 2022 - August 2022

Cloud Engineer Intern, Scientific and Cloud Computing Group

- Reduced AWS cloud vulnerabilities by 95% by creating automated scripts
- Deployed Ansible playbooks via Jenkins pipelines to run scripts and report vulnerabilities via daily emails
 - Technologies: AWS Cloud, Jenkins, GitHub, YAML, Ansible, Jira

AbbVie Inc., North Chicago, IL

June 2021 - August 2021

Cloud Engineer Intern, Scientific and Cloud Computing Group

- Decreased AWS cloud costs by 90% through migration of CI/CD pipeline orchestration tools
- Used Dockerfile to create the Jenkins masters and agents, leveraging Ansible to deploy and run commands
- Implemented Jenkins pipelines in Amazon Elastic Kubernetes Service (EKS)
 - Technologies: AWS Cloud, AWS DevOps, Jenkins, Kubernetes, YAML, Docker, Ansible, Linux, Jira

PROJECTS**Benchmarking Quantum Error Correction Decoders on the Surface Code**

February 2026 - May 2026

- Designed and implemented a simulation pipeline for testing quantum error correction (QEC) decoders on the surface code
- Evaluated different decoder performances across accuracy, latency, threshold, robustness, and scalability tests
- Analyzed results across various physical error rates and code distances to determine real world use cases for all decoders
 - Source code and research report can be found at https://github.com/tvc04/qec_decoding

SHELLS

September 2025 - December 2025

- Created the video game, SHELLS, in the Unity game engine, exhibited at public showcases, and hosted online
- Developed and designed a 1v1 party style game focused around a chaotic and synergistic upgrade system
- Collaborated with 4 teammates where I coordinated twice-weekly meetings and helped schedule and assign tasks
- Presented deliverables at multiple playtesting sessions to meet potential users and receive feedback

SKILLS**Languages:** Python, C++, C, C#, Java, JavaScript, HTML, CSS, Swift, YAML**Tools:** GitHub, LLMs, Qiskit, Streamlit, Slurm, Unity, Chrome Extensions, Web UI/UX, Pandas, Flask, AWS DevOps, AWS Cloud, Jenkins, Ansible, Docker, Kubernetes, Linux, Jira